# Unit Synthesis Review

Grade 4 Energy Transfer & Transformation — Week 9

Name: _____ Date: _____

*Use varied retrieval — not just memory. Show evidence for your ideas.*

## A. ENERGY → CHANGE → EVIDENCE

1. Energy is the ability to _____ .

2. Look at each change. Circle the BEST evidence that energy is involved.

a) A ball rolls across the floor.

b) A bell rings across the playground.

c) A plant grows taller.

3. Pick ONE change above. Write the evidence you would observe:

## B. MOTION + ENERGY OF MOTION

4. What does a moving object have that a still object does not?

5. Give ONE example of a moving object causing a change:

## C. SAME OBJECT / DIFFERENT SPEED

6. Read the table. The SAME toy car rolled down a ramp at different heights and pushed a cup.

| Ramp height | Cup movement (measured) | Energy of motion (inferred) |
|---|---|---|
| Low | moved a little | less |
| Medium | moved more | more |
| High | moved the most | most |

What do we MEASURE? What do we INFER? Circle one pair:

(A) measure cup movement, infer speed (B) measure speed, infer cup movement

7. For the SAME object, as speed goes up, energy of motion goes _____ .

## D. TRANSFER PATHS

8. Name the FOUR ways energy can transfer from place to place:

_____ _____ _____ _____

9. Warm water warms a cool spoon. What kind of energy transfers to the spoon?

## E. TRANSFER / TRANSFORMATION / BOTH

10. Label each as TRANSFER, TRANSFORMATION, or BOTH.

a) A flashlight turns on (battery → light): _____

b) Warm water warms a spoon (heat moves, stays heat): _____

c) A domino falls and clicks (motion passes + sound): _____

d) A buzzer buzzes (electrical → sound): _____

## F. COLLISION PREDICTIONS

11. A toy car rolls into a stack of blocks. The top block falls off.

You add weight to the car (change ONLY weight) and test again.

Predict what happens, and explain using energy:

12. During a collision, energy of motion transfers to the target, and some becomes _____ .

## G. DATA / GRAPH READING

13. The graph shows cup movement at different ramp heights for the SAME car.

As ramp height increases, cup movement _____ .

This is evidence that _____ .

## H. ENGINEERING — criteria / constraints / test / refine

14. Match the word to its meaning by drawing a line:

criteria — the limits (materials, time, safety)

constraints — what a good solution must do

refine — to improve based on test evidence

15. A fair test changes _____ thing(s) and keeps everything else the same.

## I. EDISON / ENGINEERING PERSISTENCE

16. Did Edison invent the FIRST light bulb? Circle: YES / NO

17. What did Edison actually do? (one sentence)

18. How many materials did Edison's team test for the filament? (about) _____

19. What does persistence mean in engineering?

## J. SHORT EXPLANATION

20. Pick ONE device (flashlight, buzzer, ramp toy, or warm-water spoon).

Describe the energy change from beginning to end. Use TRANSFER or TRANSFORMATION.

Claim: _____

Evidence: _____

Reasoning (why the evidence supports your claim): _____

*Provenance: ADAPTED (synthesis aligned to TG Unit Review pp.94-98 + official assessment concepts, re-authored).*